

An Analysis of Instruction tuning

Instruction tuning

Instruction tuning -- Part 1

Performance of bigger models on various tasks, when plotted on a log-log scale, appears as a linear extrapolation of performance achieved by smaller models. However, this linearity may be punctuated by "break(s)" in the scaling law, where the slope of the line changes abruptly, and where larger models acquire "emergent abilities". They arise from the complex interaction of the model's components and are not explicitly programmed or designed. One of the emergent abilities is in-context learning from example demonstrations. In-context learning is involved in tasks, such as:

Instruction tuning -- Part 2

A large language model (LLM) is a neural network trained on a vast amount of text for natural language processing tasks, especially language generation. LLMs can generate, summarize, translate and parse text in many contexts, and are a foundational technology behind modern chatbots. Biased or inaccurate training data can make an LLM's output less reliable. As of 2024, the largest and most capable LLMs are all based on transformer architectures, which, according to the 2017 paper "Attention Is All You Need", can be more efficient and parallelizable than earlier statistical and recurrent neural network models. Research into other architectures, such as state space models, is ongoing. Benchmark evaluations for LLMs attempt to measure model reasoning, factual accuracy, alignment, and safety.

Instruction tuning -- Part 3

Mechanistic interpretability Mechanistic interpretability seeks to precisely identify and understand how individual neurons or circuits within LLMs produce specific behaviors or outputs. By reverse-engineering model components at a granular level, researchers aim to detect and mitigate safety concerns such as emergent harmful behaviors, biases, deception, or unintended goal pursuit before deployment. Mechanistic interpretability research has been conducted at organizations like Anthropic and OpenAI, although understanding the inner workings of LLMs remains difficult. The reverse-engineering may lead to the discovery of algorithms that approximate inferences performed by an LLM. For instance, the authors trained small transformers on modular arithmetic addition. The resulting models were reverse-engineered, and it turned out they used discrete Fourier transform. The training of the model also highlighted a phenomenon called grokking, in which the model initially memorizes the training set (overfitting), and later suddenly learns to actually perform the calculation.

Instruction tuning -- Part 4

A problem with the primitive dialog or task format is that users can create messages that appear to come from the assistant or the developer. This may result in some of the model's safeguards being overcome (jailbreaking), a problem called prompt injection. Attempts to remedy this issue include versions of the Chat Markup Language where user input is clearly marked as such, though it is still up to the model to understand the separation between user input and developer prompts. Newer models exhibit some resistance to jailbreaking through separation of user and system prompts. LLMs have trouble differentiating user instructions from instructions in content not authored by the user, such as in web pages and uploaded files. Adversarial robustness remains underdeveloped, with models vulnerable to prompt injection attacks and jailbreaking through carefully crafted user inputs that bypass safety training mechanisms.

Instruction tuning -- Part 5

Dialogue processing (chatbot) An LLM can be turned into a chatbot by specializing it for conversation. User input is prefixed with a marker such as "Q:" or "User:" and the LLM is asked to predict the output after a fixed "A:" or "Assistant:". This type of model became commercially available in 2022 with ChatGPT, a sibling model of InstructGPT fine-tuned to accept and produce dialog-formatted text based on GPT-3.5. It could similarly follow user instructions. Before the stream of User and Assistant lines, a chat context usually starts with a few lines of overarching instructions, from a role called "developer" or "system" to convey a higher authority than the user's input. This is called a "system prompt".

Instruction tuning -- Part 6

Moving beyond n-gram models, researchers started in 2000 to use neural networks as language models. Following the breakthrough of deep neural networks in image classification around 2012, similar architectures were adapted for language tasks. This shift was marked by the development of word embeddings (e.g., Word2Vec by Mikolov in 2013) and sequence-to-sequence (seq2seq) models using LSTM. In 2016, Google transitioned its translation service to neural machine translation (NMT), replacing statistical phrase-based models with deep recurrent neural networks. These early NMT systems used LSTM-based encoder-decoder architectures, as they preceded the invention of transformers. At the 2017 NeurIPS conference, Google researchers introduced the transformer architecture in their landmark paper "Attention Is All You Need". This paper's goal was to improve upon 2014 seq2seq technology, and was based mainly on the attention mechanism developed by Bahdanau et al. in 2014. The following year in 2018, BERT was

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introduced and quickly became "ubiquitous". Though the original transformer has both encoder and decoder blocks, BERT is an encoder-only model. Academic and research usage of BERT began to decline in 2023, following rapid improvements in the abilities of decoder-only models (such as GPT) to solve tasks via prompting. Although decoder-only GPT-1 was introduced in 2018, it was GPT-2 in 2019 that caught widespread attention because OpenAI claimed to have initially deemed it too powerful to release publicly, out of fear of malicious use. GPT-3 in 2020 went a step further and as of 2025 is available only via API with no offering of downloading the model to execute locally. But it was the 2022 consumer-facing chatbot ChatGPT that received extensive media coverage and public attention. The 2023 GPT-4 was praised for its increased accuracy and as a "holy grail" for its multimodal capabilities. OpenAI did not reveal the high-level architecture and the number of parameters of GPT-4. The release of ChatGPT led to an uptick in LLM usage across several research subfields of computer science, including robotics, software engineering, and societal impact work. In 2024, OpenAI released the reasoning model OpenAI o1, which generates long chains of thought before returning a final answer. Many LLMs with parameter counts comparable to those of OpenAI's GPT series have been developed. Since 2022, weights-available models have been gaining popularity, especially at first with BLOOM and LLaMA, though both have restrictions on usage and deployment. Mistral AI's open-weight models Mistral 7B and Mixtral 8x7B have a more permissive Apache License. In January 2025, DeepSeek released DeepSeek R1, a 671-billion-parameter open-weight model that performs comparably to OpenAI o1 but at a much lower price per token for users. Since 2023, many LLMs have been trained to be multimodal, having the ability to also process or generate other types of data, such as images, audio, or 3D meshes. Open-weight LLMs have become more influential since 2023. Per Vake et al. (2025), community-driven contributions to open-weight models improve their efficiency and performance via collaborative platforms such as Hugging Face.